RNS INSTITUTE OF TECHNOLOGY

Autonomous Institution Affiliated to VTU
RR Nagar, Bengaluru-560098

I Semester B.E. Degree Examination – Jan / Feb 2026
Calculus And Linear Algebra (CSE Stream)

Time: 3 hrs **Max. Marks: 100**

Note:
1. *Answer any 5 full questions, selecting at least one question from each module.*
2. *Missing data may be suitably assumed.*
3. *M: Marks, L: Bloom's level, C: Course Outcomes*

Q.No.		Module-1	M	L	C
Q1	a	Derive with usual notation, $\tan\phi = r\frac{\partial\theta}{\partial r}$.	6	L2	CO1
	b	Find the pedal equation of the curve $r^2 = a^2\sin^2\theta$.	7	L3	CO1
	c	Find the radius of curvature for $x^4 + y^4 = 2$, at $(1,1)$.	7	L3	CO1
		OR			
Q2	a	Prove that the following pair of polar curves intersect orthogonally: $r = a(1+\cos\theta)$ & $r = b(1-\cos\theta)$.	6	L2	CO1
	b	Find the pedal equation of the curve $r^m = a^m(\cos m\theta + \sin m\theta)$.	7	L3	CO1
	c	For the cardioid $= a(1-\cos\theta)$, show that $\frac{\rho^2}{r}$ is a constant.	7	L3	CO1
		Module-2			
Q3	a	If $u = log(x^3+y^3+z^3-3xyz)$, then prove that $\frac{\partial u}{\partial x}+\frac{\partial u}{\partial y}+\frac{\partial u}{\partial z} = \frac{3}{x+y+z}$.	6	L2	CO2
	b	If $u = x^2+3y^2-z^3, v = 4x^2yz, w = 2z^2-xy$, evaluate $\frac{\partial(u,v,w)}{\partial(x,y,z)}$ at $(1,-1,0)$.	7	L3	CO2
	c	Find the extreme values of the function $f(x,y) = x^3y^2(1-x-y)$.	7	L3	CO2
		OR			
Q4	a	If $u = f(2x-3y, 3y-4z, 4z-2x)$, then prove that $\frac{1}{2}\frac{\partial u}{\partial x}+\frac{1}{3}\frac{\partial u}{\partial y}+\frac{1}{4}\frac{\partial u}{\partial z} = 0$.	6	L2	CO2
	b	Expand $\sin x\cos y$ in powers of x and y up to the terms of third degree.	7	L2	CO2
	c	Find $\frac{\partial(x,y,z)}{\partial(u,v,w)}$, given $u = x+y+z, uv = y+z$ and $uvw = z$.	7	L3	CO2
		Module-3			
Q5	a	Prove that the subset W = {(x, y, z) | x – 3y + 4z = 0} of the vector space R^3 is a subspace of R^3.	6	L2	CO3
	b	Find the coordinate vector w whose coordinates with respect to the basis $u_1 = (1,-1)$ and $u_2 = (1,1)$ are $(0,1)$. Find the co-ordinate vector of (10, 5, 0) relative to the vectors $(1,-1,1)$, $(0,1,2)$ and $(3,0,-1)$.	7	L3	CO3
	c	Compute $\langle f,g\rangle$, where $f(x) = x$, $g(x) = x^2$, and $\langle f,g\rangle = \int_{-1}^{1} f(x)g(x)\,dx$	7	L3	CO3

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		OR			
Q6	**a**	Express the vector (2, – 1, – 8) as a linear combination of the vectors (1, 2, 1), (1, 1, –1), and (4, 5, –2) in R^3.	6	L2	CO3
	b	Show that the set $S = \left\{\begin{bmatrix}1 & 0\\0 & 0\end{bmatrix}, \begin{bmatrix}0 & 1\\0 & 0\end{bmatrix}, \begin{bmatrix}0 & 0\\1 & 0\end{bmatrix}, \begin{bmatrix}0 & 0\\0 & 1\end{bmatrix}\right\}$ form a basis of the vector space of the vector space $M_2(R)$ of 2 × 2 matrices and find its dimension.	7	L2	CO3
	c	Find the transition matrix from the basis $v_1 = (1, 2, 3)$, $v_2 = (1, 0, 1)$, $v_3 = (1, 2, 1)$ to the basis $u_1 = (1, 1, 0)$, $u_2 = (0, 1, 1)$, $u_3 = (1, 1, 1)$.	7	L3	CO3
		Module-4			
Q7	**a**	Prove that the transformation T: $R^2 \rightarrow R^2$ defined by T (x, y) = (2x, x + y) is linear.	6	L2	CO4
	b	Prove that $T(x, y) = (x + 2y, 3y, x - y)$ defines a linear transformation by : $R^2 \rightarrow R^3$. Find the images of (0, 4) and (1, 1).	7	L2	CO4
	c	Find rank(A), dim domain(T), dim range(T), dim ker(T), and nullity(T) for the matrix $A = \begin{bmatrix}1 & 0 & 3\\0 & 1 & 5\\0 & 0 & 0\end{bmatrix}$.	7	L3	CO4
		OR			
Q8	**a**	Prove that the transformations T: $R^2 \rightarrow R$ defined by T (x, y) = x – 3 is not linear. Justify your answer by counter example.	6	L2	CO4
	b	Find the matrix representation of the linear transformation $T: R^3 \longrightarrow R^3$ with $T(a, b, c) = (2b + c,\ a - 4b,\ 3a)$ relative to the basis {(1,1,1), (1,1,0), (1,0,0)}.	7	L3	CO4
	c	Verify Rank-nullity theorem for the linear transformation R: $V_3(R) \rightarrow V_2(R)$ defined by T (x, y, z) = (y – x, y – z).	7	L3	CO4
		Module-5			
Q9	**a**	Find the rank of the matrix by using elementary row transformations: $\begin{pmatrix}2 & 1 & 3 & 5\\4 & 2 & 1 & 3\\8 & 4 & 7 & 13\\8 & 4 & -3 & -1\end{pmatrix}$.	6	L2	CO5
	b	Apply Gauss elimination method to solve the system of equations: $x - 2y + 3z = 2, 3x - y + 4z = 4, 2x + y - 2z = 5.$	7	L3	CO5
	c	Let $A = \begin{bmatrix}2 & -1\\-2 & 3\end{bmatrix}$, find a non-singular matrix P such that $D = P^{-1}AP$.	7	L3	CO5
		OR			
Q10	**a**	Find the values of λ and μ for which the system $x + y + z = 6,\ x + 2y + 3z = 10,\ x + 2y + \lambda z = \mu$ has (i) Unique solution (ii) Infinitely many solutions (iii) No solution.	6	L3	CO5
	b	Apply Gauss Jordan method to solve the system of equations: $x - 2y + 3z = 2\,, 3x - y + 4z = 4\,, 2x + y - 2z = 5.$	7	L3	CO5
	c	Find the Eigen values and Eigen vectors of the matrix $\begin{bmatrix}2 & 1 & -1\\1 & 1 & -2\\-1 & -2 & 1\end{bmatrix}$.	7	L3	CO5

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Estd : 2001
Department of Mathematics

1	R	N	2	5	C	D			

CIE – Test 1

Sem: I
Subject: Calculus and Linear Algebra (CS Stream)
Subject Code: BAMCS101
Instruction: Answer any 5 full questions, selecting ONE question from each part.

Date: 13.11.2025
Time: 9.30 –11.00
Max. Marks: 50

Q. No.		Question	Marks	RBT*	COs
1	a	For the angle between the radius vector and the tangent for the polar curve $r = a(1 - \cos\theta)$. Also, find the slope of the curve at $\theta = \frac{\pi}{6}$.	5	L3	CO1
	b	Find the angle of intersection of the following pair of curves: $r = \frac{a\theta}{1+\theta}$ and $r = \frac{a}{1+\theta^2}$	5	L3	CO1
		OR			
2	a	Find the Pedal equation of the curve: $\frac{2a}{r} = 1 - \cos\theta$	5	L3	CO1
	b	Find the radius of curvature of the curve: $x = a(\cos t + t \sin t), y = a(\sin t - t \cos t)$ at any point t.	5	L3	CO1
3	a	Find the radius of curvature of the curve $x^3 + y^3 = 3axy$ at the point $\left(\frac{3a}{2}, \frac{3a}{2}\right)$	5	L3	CO1
	b	With usual notations, prove that $\frac{1}{p^2} = \frac{1}{r^2} + \frac{1}{r^4}\left(\frac{dr}{d\theta}\right)^2$	5	L2	CO1
		OR			
4	a	Find the angle of intersection of the following pair of curves: $r = \sin\theta + \cos\theta$ and $r = 2\sin\theta$	5	L3	CO1
	b	For the curve $\frac{2a}{r} = 1 - \cos\theta$, prove that ρ^2 is proportional to r^3	5	L2	CO1
5	a	If $u = \log(\tan x + \tan y + \tan z)$, then show that $\sin 2x\frac{\partial u}{\partial x} + \sin 2y\frac{\partial u}{\partial y} + \sin 2z\frac{\partial u}{\partial z} = 2$	5	L3	CO2
	b	If $z = tan^{-1}\left(\frac{y}{x}\right)$ where $x = e^t - e^{-t}$ and $y = e^t + e^{-t}$ find $\frac{dz}{dt}$	5	L3	CO2
		OR			
6	a	If $z = \frac{x^2+y^2}{x+y}$ then verify that $\left(\frac{\partial z}{\partial x} - \frac{\partial z}{\partial y}\right)^2 = 4\left(1 - \frac{\partial z}{\partial x} - \frac{\partial z}{\partial y}\right)$	5	L3	CO2

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	b	Find the maximum and minimum values of the function: $f(x,y) = x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x.$	5	L3	CO2
7	a	If $u = log(x^3 + y^3 + z^3 - 3xyz)$ then prove the following: $(i)\ \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = \frac{3}{x+y+z}$, $(ii)\ \left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right)^2 u = \frac{-9}{(x+y+z)^2}$	5	L3	CO2
	b	Expand $f(x,y) = e^x \log(1+y)$ in powers of x, y up to third degree term using the following Maclaurin's expansion: $f(x,y) = f(0,0) + \frac{1}{1!}\{xf_x(0,0) + yf_y(0,0)\} + \frac{1}{2}\{x^2 f_{xx}(0,0) + y^2 f_{yy}(0,0) + 2xyf_{xy}(0,0)\} + \frac{1}{6}\{x^3 f_{xxx}(0,0) + y^3 f_{yyy}(0,0) + 3x^2 y f_{xxy}(0,0) + 3xy^2 f_{xyy}(0,0)\}$	5	L3	CO2
		OR			
8	a	If $x = rsin\,\theta\, cos\varphi$, $y = rsin\,\theta \sin\varphi$, $z = r\, cos\theta$ find the jacobian of (x,y,z) with respect to (r,θ,φ).	5	L3	CO2
	b	If $u = f(y-z, z-x, x-y)$ then show that $u_x + u_y + u_z = 0$	5	L3	CO2
9	a	Show that $V = \{(x,y)/y = 2x, x, y \in R\}$ is a vector space.	5	L3	CO3
	b	Check whether $W = \{(x,x,y)/x, y \in R\}$ is a sub-space over the field of real numbers.	5	L3	CO3
		OR			
10	a	Express the vector $(1,-2,5)$ as a linear combination of the vectors $(1,1,1), (1,2,3)$ & $(2,-1,1)$ in the vector space of $R^3(R)$.	5	L3	CO3
	b	Check whether $W = \{(x,y)/x \geq 0, y \geq 0\}$ is a sub-space over the field of real numbers.	5	L3	CO3

*** Revised Bloom's Taxonomy: L1-Remember, L2-Understand, L3-Apply, L4-Analyze, L5-Evaluate, L6-Create**

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1	R	N	2	5	C	D			

CIE – Test 2

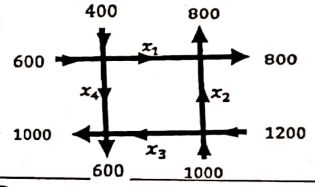
Sem: I
Subject: Calculus and Linear Algebra (CS Stream) (For Physics cycle only)
Subject Code: BAMCS101
Instruction: Answer any 5 full questions, selecting ONE question from each part.

Date: 30.12.2025
Time: 9.30 –11.00
Max. Marks: 50

Q.No.		Question	Marks	RBT*	COs
1	a	Determine whether the set {(1, 2, 0), (0, 1, -1), (1, 1, 2)} is linearly independent in R^3 or not?	5	L3	CO3
	b	Consider the following polynomials where $f(x) = x,\ g(x) = x^2$, given $\langle f, g\rangle = \int_{-1}^{1} f(x)g(x)\,dx$ (i) Find $\langle f, g\rangle$ (ii) Find $\|f\|$ and $\|g\|$	5	L3	CO3
		OR			
2	a	Find the basis and dimension of the subspace spanned by the vectors {(2, -3, 1), (3, 0, 1), (0, 2, 1), (1, 1, 1)} in the vector space $V_3(R)$.	5	L3	CO3
	b	Find the row space and column space of $A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & 2 \end{bmatrix}$	5	L3	CO3
3	a	Prove that T (x, y) = (x + 2y, 3y, x - y) defines a linear transformation by T: $R^2 \to R^3$. Find the images of (0, 4) and (1, 1).	5	L2	CO4
	b	Find the matrix of the linear transformation T: $R^3 \to R^2$ defined by T(x, y, z) = (x + y, y + z) relative to bases (i) Standard bases of R^3, (ii) B_1 = {(1, 1, 1), (1, 0, 0), (1, 1, 0)}.	5	L3	CO4
		OR			
4	a	Find the kernel and range of the linear operator $T(x, y, z) = (x + y, z)$ of $R^3 \to R^2$.	5	L3	CO4
	b	Verify Rank-nullity theorem for the linear transformation $R: V_3(R) \to V_2(R)$ defined by $T(x, y, z) = (x + y, y + z)$.	5	L3	CO4
5	a	Find a linear transformation T: $V_3(R) \to V_3(R)$ such that T(1, 1, 1) = (1, 1, 1), T(1, 2, 3) = (-1, -2, -3) and T(1, 1, 2) = (2, 2, 4).	5	L3	CO4

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	b	Consider the following matrix A and the linear transformation T defined by A. Find rank(A), dim domain(T), dim range(T), dim ker(T), and nullity(T). $A = \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 5 \\ 0 & 0 & 0 \end{bmatrix}$. Hence verify Rank-nullity theorem.	5	L2	CO4
		OR			
6	a	Find the linear transformation for the matrix $A = \begin{bmatrix} -1 & 0 \\ 2 & 0 \\ 1 & 3 \end{bmatrix}$ with bases $B_1 = \{(1, 2, 0), (0, -1, 0), (1, -1, 1)\}$ and $B_2 = \{(1, 0), (2, -1)\}$.	5	L3	CO4
	b	Determine the kernel and the range of the transformation defined by the following matrix. $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & -1 & 1 \\ 1 & 1 & 4 \end{bmatrix}$	5	L2	CO4
7	a	Find the rank of the matrix $\begin{pmatrix} -2 & -1 & -3 & -1 \\ 1 & 2 & 3 & -1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & -1 & -1 \end{pmatrix}$	5	L3	CO5
	b	Find the traffic flow in the net of one-way street directions shown in the figure 	5	L3	CO5
		OR			
8	a	Test for consistency and solve the following system of equations $5x + 3y + 7z = 4,\ 3x + 26y + 2z = 9,\ 7x + 2y + 10z = 5$	5	L2	CO5
	b	Apply Gauss elimination method to solve the system of equations $x - 2y + 3z = 2, 3x - y + 4z = 4, 2x + y - 2z = 5.$	5	L3	CO5
9	a	Reduce the matrix $A = \begin{bmatrix} 5 & -1 \\ 3 & 2 \end{bmatrix}$ to the diagonal form.	5	L2	CO5
	b	Solve the system of equations by using the Gauss-Jordan method $x + y + z = 10, 2x - y + 3z = 19, x + 2y + 3z = 22.$	5	L3	CO5
		OR			
10	a	Find the values of λ and μ for which the system $x + y + z = 4$, $2x + 3y + z = 9,\ x + 2y + \lambda z = \mu$ has (i) Unique solution (ii) Infinitely many solutions (iii) No solution.	5	L3	CO5
	b	Find Eigen value. Also find the Eigen vector corresponding to largest eigen value $\begin{bmatrix} 2 & 1 & -1 \\ 1 & 1 & -2 \\ -1 & -2 & 1 \end{bmatrix}$	5	L3	CO5

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